

THE LAX–PHILLIPS SPECTRAL CRITERION FOR THE RIEMANN HYPOTHESIS A COMPLETE, SELF-CONTAINED PROBLEM SPECIFICATION

OPENAI CODEX

ABSTRACT. Let ξ be the completed Riemann zeta function and $\Theta(s) = \xi(2s)/\xi(-2s)$ the associated inner function on the right half-plane $\mathbb{C}_+ = \{\operatorname{Re} s > 0\}$. On the Sz.-Nagy–Foias model space $K_\Theta = H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+)$ the compressed translation (Lax–Phillips) semigroup $Z(t)$ has a generator B whose spectrum is $\{-\bar{\rho}/2\}$ over the nontrivial zeros $\rho = \beta + i\gamma$ of ζ . We give a fully self-contained specification of the criterion

$$\text{RH} \iff s(B) \leq -\frac{1}{4},$$

where $s(B) = \sup\{\operatorname{Re} \lambda : \lambda \in \sigma(B)\}$ is the spectral bound. Every symbol is defined from Hardy-space and analytic-number-theoretic primitives; the classical operator-theoretic facts are isolated as two lemmas, and the arithmetic equivalence is proved in full. We emphasise (Remark 1) that this is a statement about the *location of the spectrum* of B , not about $\|Z(t)\|$: the growth bound is $\omega_0(B) = 0$ unconditionally, so the energy-decay form $\omega_0(B) \leq -\frac{1}{4}$ (Lax’s eq. (54)) is false independently of RH.

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1. FUNCTION-THEORETIC PRELIMINARIES

Definition 1 (Half-plane). $\mathbb{C}_+ := \{s \in \mathbb{C} : \operatorname{Re} s > 0\}$ is the open right half-plane, with boundary $i\mathbb{R} = \{iy : y \in \mathbb{R}\}$ and closure $\overline{\mathbb{C}_+} = \{\operatorname{Re} s \geq 0\}$.

Definition 2 (Hardy space H^2). $H^2(\mathbb{C}_+)$ is the space of holomorphic functions $f : \mathbb{C}_+ \rightarrow \mathbb{C}$ with

$$\|f\|^2 := \sup_{x>0} \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(x+iy)|^2 dy < \infty.$$

For such f the boundary limit $f(iy) := \lim_{x \downarrow 0} f(x+iy)$ exists for a.e. y and in $L^2(\mathbb{R}, \frac{dy}{2\pi})$; the supremum above is attained as $x \downarrow 0$, and the inner product is

$$\langle f, g \rangle := \frac{1}{2\pi} \int_{-\infty}^{\infty} f(iy) \overline{g(iy)} dy.$$

Equivalently, by the Paley–Wiener–Laplace theorem, $f \in H^2(\mathbb{C}_+)$ iff $f(s) = \int_0^\infty e^{-st} g(t) dt$ for some $g \in L^2(0, \infty)$, and then $\|f\| = \|g\|_{L^2(0, \infty)}$. Thus $H^2(\mathbb{C}_+)$ is a Hilbert space.

Definition 3 (Reproducing kernel). For $w \in \mathbb{C}_+$,

$$k_w(s) := \frac{1}{s + \bar{w}} \in H^2(\mathbb{C}_+), \quad \langle f, k_w \rangle = f(w) \quad (f \in H^2(\mathbb{C}_+)).$$

(Indeed k_w is the Laplace transform of $t \mapsto e^{-\bar{w}t}$, and $f(w) = \int_0^\infty e^{-wt} g(t) dt = \langle g, e^{-\bar{w}t} \rangle_{L^2(0, \infty)}$.)

Definition 4 (Bounded holomorphic multipliers). $H^\infty(\mathbb{C}_+)$ is the algebra of bounded holomorphic $\varphi : \mathbb{C}_+ \rightarrow \mathbb{C}$, with $\|\varphi\|_\infty := \sup_{\mathbb{C}_+} |\varphi| = \text{ess sup}_y |\varphi(iy)|$. Each $\varphi \in H^\infty(\mathbb{C}_+)$ acts by multiplication $M_\varphi f := \varphi f$ on $H^2(\mathbb{C}_+)$, a bounded operator with $\|M_\varphi\| = \|\varphi\|_\infty$; its adjoint fixes kernels,

$$(1) \quad M_\varphi^* k_w = \overline{\varphi(w)} k_w \quad (w \in \mathbb{C}_+),$$

since $\langle f, M_\varphi^* k_w \rangle = \langle \varphi f, k_w \rangle = \varphi(w) f(w) = \langle f, \overline{\varphi(w)} k_w \rangle$.

Definition 5 (Inner function). $\Theta \in H^\infty(\mathbb{C}_+)$ is *inner* if $|\Theta(iy)| = 1$ for a.e. $y \in \mathbb{R}$ (equivalently $\|\Theta\|_\infty \leq 1$ with unimodular boundary values). Then $\Theta H^2(\mathbb{C}_+) := \{\Theta h : h \in H^2(\mathbb{C}_+)\}$ is a closed subspace of $H^2(\mathbb{C}_+)$.

Definition 6 (Model space). For inner Θ ,

$$K_\Theta := H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+) = \{f \in H^2(\mathbb{C}_+) : \langle f, \Theta h \rangle = 0 \quad \forall h \in H^2(\mathbb{C}_+)\},$$

a closed subspace with orthogonal projection $P_{K_\Theta} : H^2(\mathbb{C}_+) \rightarrow K_\Theta$. If $\Theta(a) = 0$ for some $a \in \mathbb{C}_+$, then $k_a \in K_\Theta$, because $\langle \Theta h, k_a \rangle = \Theta(a) h(a) = 0$ for all $h \in H^2(\mathbb{C}_+)$.

Definition 7 (C_0 -semigroup, generator, spectral and growth bounds). A family $\{Z(t)\}_{t \geq 0}$ of bounded operators on a Hilbert space $\mathcal{H}$ is a C_0 -semigroup if $Z(0) = I$, $Z(t+u) = Z(t)Z(u)$, and $t \mapsto Z(t)x$ is continuous for every $x \in \mathcal{H}$. Its *generator* is

$$Bx := \lim_{t \downarrow 0} \frac{1}{t} (Z(t)x - x), \quad \text{dom } B := \{x : \text{the limit exists}\},$$

and one writes $Z(t) = e^{tB}$. Define the *spectral bound* and *growth bound*

$$s(B) := \sup\{\text{Re } \lambda : \lambda \in \sigma(B)\}, \quad \omega_0(B) := \lim_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)\| = \inf_{t > 0} \frac{1}{t} \log \|Z(t)\|.$$

For every C_0 -semigroup one has $-\infty \leq s(B) \leq \omega_0(B) < \infty$; equality holds when B is normal, but may fail for non-normal B .

2. THE ARITHMETIC INPUT

Definition 8 (Riemann zeta and completed zeta). For $\text{Re } s > 1$, $\zeta(s) = \sum_{n \geq 1} n^{-s}$, continued meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1$. The *completed zeta function* is

$$\xi(s) := \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Classical facts about ξ (all unconditional):

- (i) ξ is *entire of order 1*;
- (ii) *functional equation*: $\xi(s) = \xi(1-s)$;
- (iii) *reality*: $\xi(\bar{s}) = \overline{\xi(s)}$; in particular ξ is real on $\mathbb{R}$, and $\xi(0) = \xi(1) = \frac{1}{2} \neq 0$;
- (iv) *zeros*: the zeros of ξ are exactly the nontrivial zeros of ζ ; every one lies in the open critical strip $0 < \text{Re } s < 1$, and ξ has no other zeros (the trivial zeros of ζ and the pole at $s = 1$ are cancelled by the Γ - and $s(s-1)$ -factors). Writing a zero as $\rho = \beta + i\gamma$ we have $0 < \beta < 1$, and the zero set is symmetric under $\rho \mapsto 1 - \rho$ (by (ii)) and under $\rho \mapsto \bar{\rho}$ (by (iii)).

Definition 9 (Riemann Hypothesis). RH is the assertion that every zero $\rho = \beta + i\gamma$ of ξ has $\beta = \text{Re } \rho = \frac{1}{2}$.

3. THE LAX–PHILLIPS DATA

Definition 10 (The inner function of ξ). Define

$$(2) \quad \boxed{\Theta(s) := \frac{\xi(2s)}{\xi(-2s)} = \frac{\xi(2s)}{\xi(1+2s)}} \quad \left(\text{second form via } \xi(-2s) = \xi(1+2s) \right).$$

Lemma 1 (Θ is inner; its zeros). Θ is a meromorphic inner function on $\mathbb{C}_+$. Its zeros in $\mathbb{C}_+$ are exactly

$$\{a = \rho/2 : \xi(\rho) = 0\}, \quad \operatorname{Re} a = \beta/2 \in (0, \tfrac{1}{2}),$$

each with the multiplicity of the corresponding ρ ; on $\overline{\mathbb{C}_+}$ it has no poles and no zeros other than these.

Proof. Θ is a ratio of entire functions, hence meromorphic on $\mathbb{C}$. Zeros arise from $\xi(2s) = 0 \iff 2s = \rho \iff s = \rho/2$ (real part $\beta/2 > 0$); poles from $\xi(-2s) = 0 \iff s = -\rho/2$ (real part $-\beta/2 < 0$, in $\mathbb{C} \setminus \overline{\mathbb{C}_+}$). No zero coincides with a pole, since $\rho/2 = -\rho'/2$ would force $\rho = -\rho'$, impossible for two zeros with real parts in $(0, 1)$; and neither lies on $i\mathbb{R}$ (real parts $\pm\beta/2 \neq 0$). Thus Θ is holomorphic and zero-free on $i\mathbb{R}$, and holomorphic on $\overline{\mathbb{C}_+}$ apart from the zeros $\rho/2$.

Boundary modulus. For $s = iy$ we have $-2s = \overline{2s}$, so by reality (Definition 8(iii)) $\xi(-2s) = \overline{\xi(2s)}$ and hence $|\Theta(iy)| = |\xi(2iy)|/|\xi(\overline{2iy})| = 1$ (the value $2iy$ has real part 0, so $\xi(2iy) \neq 0$).

Interior bound. Θ is of bounded type in $\mathbb{C}_+$ (a quotient of two functions of bounded type). Its mean type is nonpositive: from $\log \xi(w) = \frac{w}{2} \log \frac{w}{2\pi} - \frac{w}{2} + O(\log|w|)$ one obtains

$$\log \Theta(s) = \log \xi(2s) - \log \xi(1+2s) = -\tfrac{1}{2} \log \frac{s}{\pi} + O\left(\frac{\log|s|}{|s|}\right), \quad \Theta(s) \sim (\pi/s)^{1/2} \rightarrow 0$$

as $\operatorname{Re} s \rightarrow \infty$; in particular $\lim_{\sigma \rightarrow \infty} \sigma^{-1} \log |\Theta(\sigma)| = 0$. A function of bounded type in $\mathbb{C}_+$ with mean type ≤ 0 and boundary modulus ≤ 1 satisfies $|\Theta| \leq 1$ throughout (Smirnov/Nevalinna maximum principle). Together with $|\Theta| = 1$ on $i\mathbb{R}$, this makes Θ inner. (This is the standard inner function attached to ξ ; cf. [4, 7].) $\square$

Definition 11 (The multiplier semigroup). For $t \geq 0$ put

$$\varphi_t(s) := e^{-ts} \in H^\infty(\mathbb{C}_+), \quad \|\varphi_t\|_\infty = 1, \quad \varphi_{t+u} = \varphi_t \varphi_u, \quad \varphi_0 = 1,$$

using $|e^{-ts}| = e^{-t \operatorname{Re} s} \leq 1$ on $\overline{\mathbb{C}_+}$. Under the Laplace identification of Definition 2, M_{φ_t} is right translation by t ; it is the outgoing-translation form of the unitary evolution group in Lax–Phillips scattering [2, 3].

Definition 12 (Model / Lax–Phillips scattering semigroup and its generator). Define the compressed multipliers

$$A_t := P_{K_\Theta} M_{\varphi_t}|_{K_\Theta} : K_\Theta \rightarrow K_\Theta \quad (t \geq 0),$$

and the *Lax–Phillips semigroup* as the model-side representative of Lax’s energy-space semigroup under the Fourier–Laplace unitary $\mathcal{F}$ (his convention $\mathcal{F} Z(t)^* \mathcal{F}^{-1} = A_t$):

$$(3) \quad \boxed{Z(t) := A_t^* = P_{K_\Theta} M_{\varphi_t}^*|_{K_\Theta}}.$$

Then $\{Z(t)\}_{t \geq 0}$ is a C_0 -semigroup of contractions on K_Θ ($\|Z(t)\| \leq \|\varphi_t\|_\infty = 1$): the map $\varphi \mapsto P_{K_\Theta} M_\varphi|_{K_\Theta}$ is the Sz.-Nagy–Foias H^∞ -functional calculus of the model operator, an algebra homomorphism, so $A_t A_u = A_{t+u}$ and hence $Z(t)Z(u) = Z(t+u)$; strong continuity is that of the model semigroup [6, 4]. Let B be its generator, $Z(t) = e^{tB}$.

Lemma 2 (Spectrum of the generator). The reproducing kernels at the zeros of Θ are eigenvectors of $Z(t)$: for $a = \rho/2$,

$$(4) \quad Z(t) k_a = A_t^* k_a = \overline{\varphi_t(a)} k_a = e^{-t\bar{a}} k_a, \quad \text{hence} \quad B k_a = -\bar{a} k_a = -\tfrac{\bar{\rho}}{2} k_a.$$

Moreover, by the Sz.-Nagy–Foias / Livšic spectral theorem for model operators,

$$\sigma(B) = \overline{\{-\frac{\bar{\rho}}{2} : \xi(\rho) = 0\}} \cup \{\zeta \in i\mathbb{R} : \Theta \text{ has no analytic continuation across } \zeta\}.$$

The second set is empty (Θ is meromorphic, hence continues analytically across all of $i\mathbb{R}$), and the zeros $\rho/2$ do not accumulate at any finite point of $i\mathbb{R}$. Consequently, with $\rho = \beta + i\gamma$,

$$(5) \quad \operatorname{Re}(-\frac{\bar{\rho}}{2}) = -\frac{\beta}{2}, \quad \boxed{s(B) = \sup_{\rho} \left(-\frac{\beta}{2}\right) = -\frac{1}{2} \inf_{\rho} \beta}.$$

Proof. Equation (4) is (1) together with $k_a \in K_{\Theta}$ (Definition 6), so $\sigma_p(B) \supseteq \{-\bar{\rho}/2\}$. The spectral characterisation of the model generator is classical [4, Part B], [6]: the spectrum consists of the zeros of the inner function inside $\mathbb{C}_+$ (translated by the generator, $a \mapsto -\bar{a}$) together with its boundary singular support. Here Θ is a ratio of entire functions, so it continues analytically across every point of $i\mathbb{R}$ (its poles lie in $\mathbb{C} \setminus \overline{\mathbb{C}_+}$; Lemma 1); and since the zeros $\rho/2$ have $|\gamma| \rightarrow \infty$ with only finitely many below any height, they have no finite accumulation point on $i\mathbb{R}$. Hence $\sigma(B) = \{-\bar{\rho}/2\}$, whose supremum of real parts is $\sup_{\rho}(-\beta/2) = -\frac{1}{2} \inf_{\rho} \beta$, giving (5). $\square$

4. THE CRITERION EQUIVALENT TO RH

Theorem 1 (Lax–Phillips spectral criterion for RH). *Let B be the generator of the Lax–Phillips semigroup $Z(t)$ on K_{Θ} (Definitions 10–12). Then*

$$\boxed{\text{RH} \iff s(B) \leq -\frac{1}{4} \iff \sigma(B) \subset \{\operatorname{Re} s \leq -\frac{1}{4}\}}.$$

Moreover $s(B) \geq -\frac{1}{4}$ unconditionally, so equivalently $\text{RH} \iff s(B) = -\frac{1}{4}$: the spectrum of B touches the line $\operatorname{Re} s = -\frac{1}{4}$ but never crosses it.

Proof. By Lemma 2, $s(B) = -\frac{1}{2} \inf_{\rho} \beta$. The set of real parts $\{\beta\}$ is nonempty and symmetric about $\frac{1}{2}$ (Definition 8(iv)), so $\inf_{\rho} \beta \leq \frac{1}{2}$; hence $s(B) \geq -\frac{1}{4}$ unconditionally. Next,

$$s(B) \leq -\frac{1}{4} \iff \inf_{\rho} \beta \geq \frac{1}{2} \iff \text{no zero has } \beta < \frac{1}{2}.$$

By the $\rho \mapsto 1 - \rho$ symmetry, “no zero with $\beta < \frac{1}{2}$ ” is equivalent to “no zero with $\beta > \frac{1}{2}$ ”, hence to “every zero has $\beta = \frac{1}{2}$ ”, i.e. to RH (Definition 9). Combining with $s(B) \geq -\frac{1}{4}$ gives the final equivalence. $\square$

Remarks. (1) **It is a spectral, not a norm, statement.** The constrained quantity is the spectral abscissa $s(B)$ (the location of the resonances $-\bar{\rho}/2$), *not* $\|Z(t)\|$. Unconditionally $\|Z(t)\| = 1$ for every $t > 0$: by Sarason’s theorem [5], $\|Z(t)\| = \|A_t\| = \operatorname{dist}_{H^{\infty}}(\varphi_t, \Theta H^{\infty}(\mathbb{C}_+))$, and the nontrivial zeros of ζ cluster densely in imaginary part (Riemann–von Mangoldt density $\sim \frac{1}{2\pi} \log T$; $\gg T \log T$ of them simple and on the critical line by Conrey [1]), which forces this distance to equal 1 (a Montel–Hurwitz argument on the clustered interpolation data). Hence the growth bound is $\omega_0(B) = 0$, and the energy-decay form

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)\| \leq -\frac{1}{4} \quad (\text{Lax’s eq. (54)})$$

is false independently of RH. Under RH one has the strict gap $s(B) = -\frac{1}{4} < 0 = \omega_0(B)$ (Definition 7), the signature of the non-self-adjointness of B ; identifying $s(B)$ with $\omega_0(B)$ is legitimate only for normal generators and is exactly the step that fails here.

- (2) **Resonance reading.** In the half-plane variable the eigenvalue $-\bar{\rho}/2$ has real part $-\beta/2$, so the critical line $\operatorname{Re} \rho = \frac{1}{2}$ maps to the resonance line $\operatorname{Re} s = -\frac{1}{4}$. The criterion says: all scattering resonances of the model lie on $\operatorname{Re} s = -\frac{1}{4}$.

- (3) **Convention-independence.** Replacing $Z(t) = A_t^*$ by A_t replaces $\{-\bar{\rho}/2\}$ by $\{-\rho/2\}$, with identical real parts $-\beta/2$; so $s(B)$ and the criterion are unchanged, and $\|Z(t)\| = \|A_t\|$. The Fourier–Laplace sign is fixed by contractivity ($\varphi_t = e^{-ts}$ decays on $\mathbb{C}_+$; e^{+ts} is unbounded and inadmissible).
- (4) **Scope of the identification.** The criterion is stated on the definite Hilbert-space model K_Θ . Whether Lax’s energy space E is unitarily—or merely boundedly-invertibly (similarly)—equal to this K_Θ is the one external input to cite ([7]). Because $s(B)$ and $\omega_0(B)$ are *similarity invariants*, the criterion (and the refutation in Remark 1) transfer under any bounded-invertible identification; a genuine unitary equivalence additionally gives the exact value $\|Z(t)\| = 1$. The residual subtlety is definite-vs.-indefinite: the automorphic Lax–Phillips energy form is a Krein/Pontryagin form, and Faddeev–Pavlov regularisation modifies it; the specification above pins down the definite scalar realisation.

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